

Graphical Method

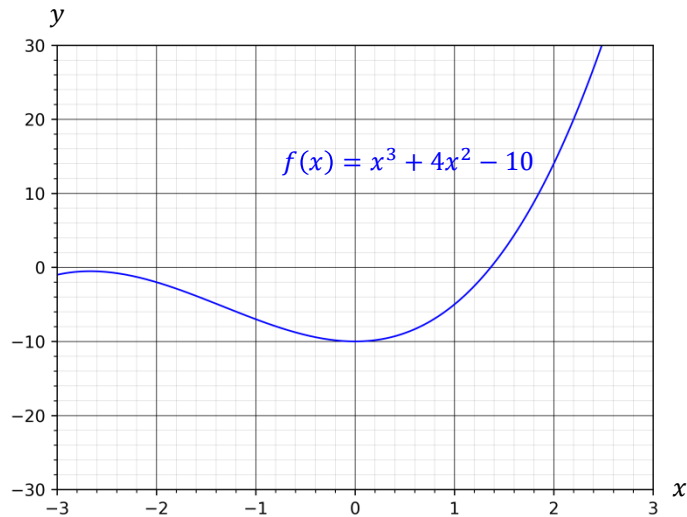
Given: $x^3 + 4x^2 = 10 \rightarrow x^3 + 4x^2 - 10 = 0$

$$f(x) = x^3 + 4x^2 - 10 = 0$$

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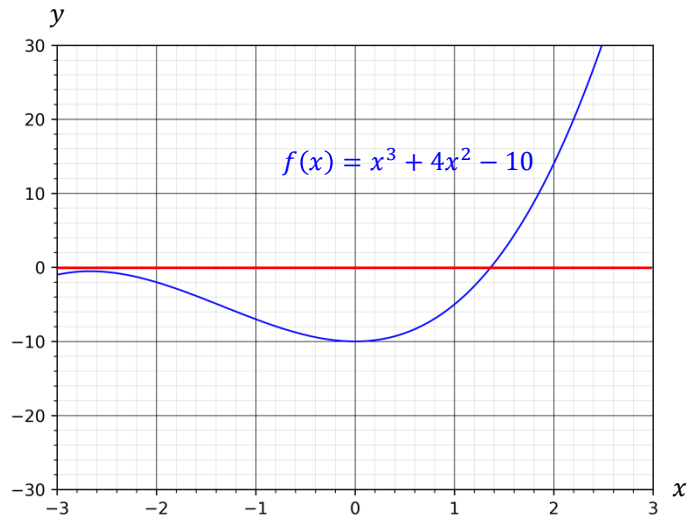
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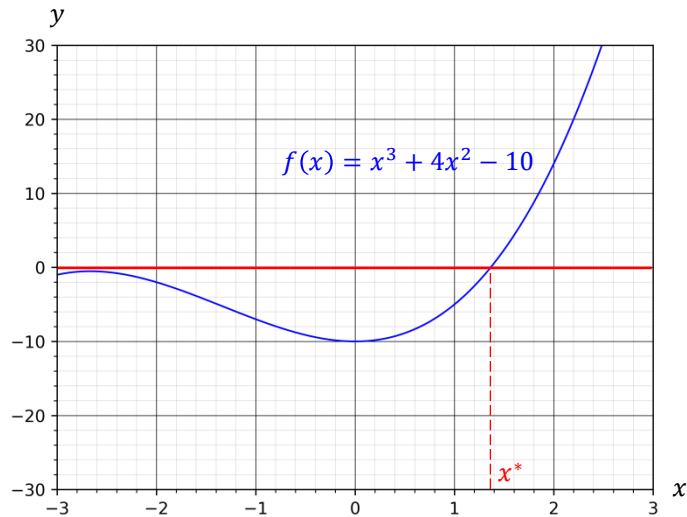
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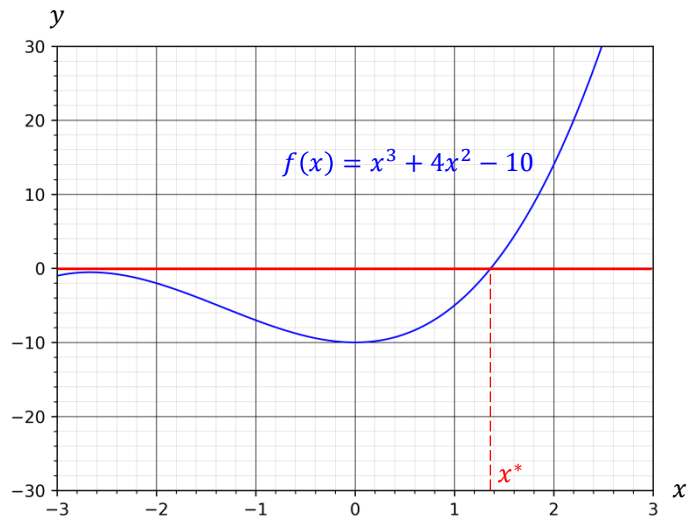
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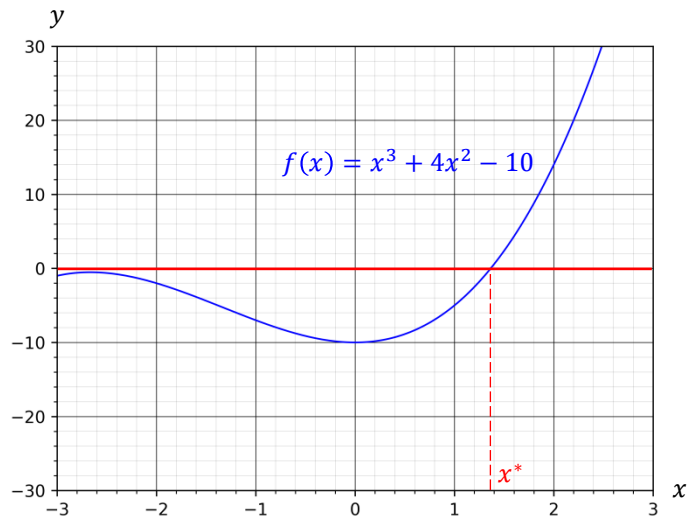
$$f(x) = x^3 + 4x^2 - 10 = 0 \rightarrow \begin{aligned} f_1(x) &= f_2(x) \\ x^3 &= 10 - 4x^2 \end{aligned}$$



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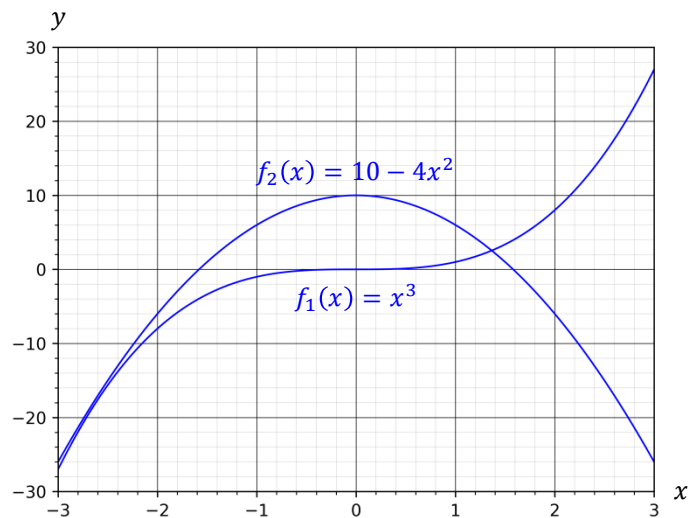
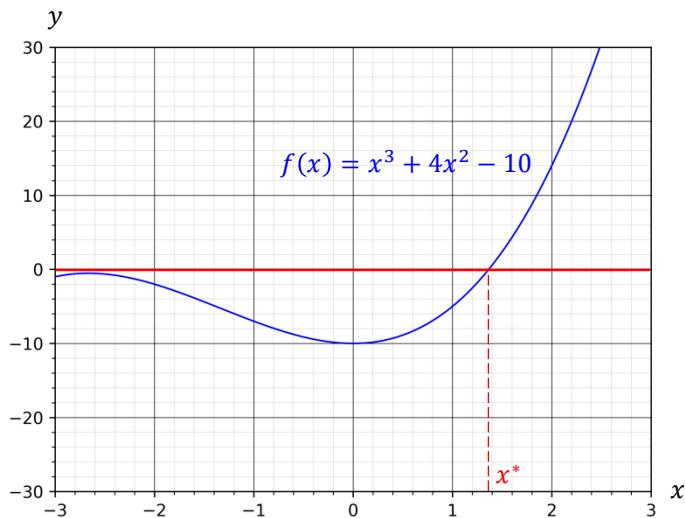
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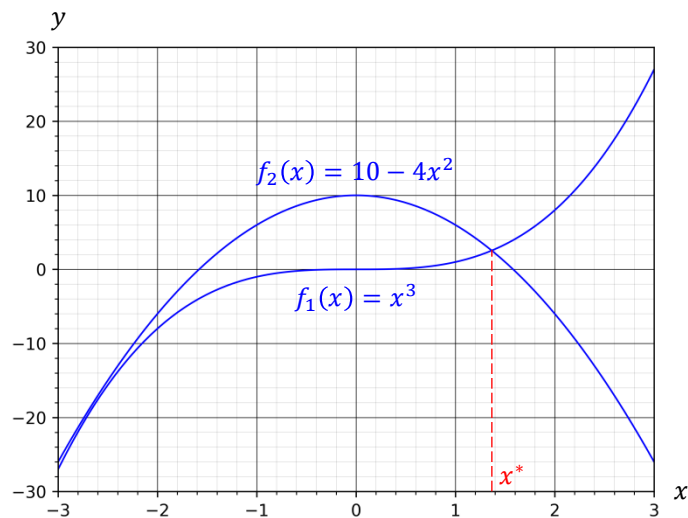
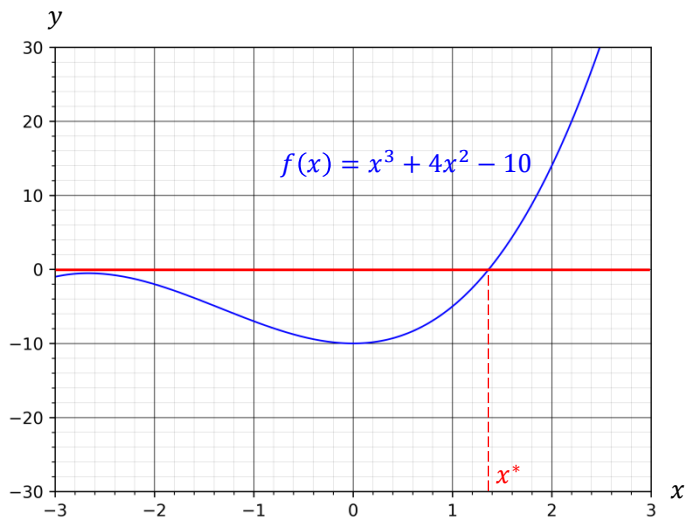
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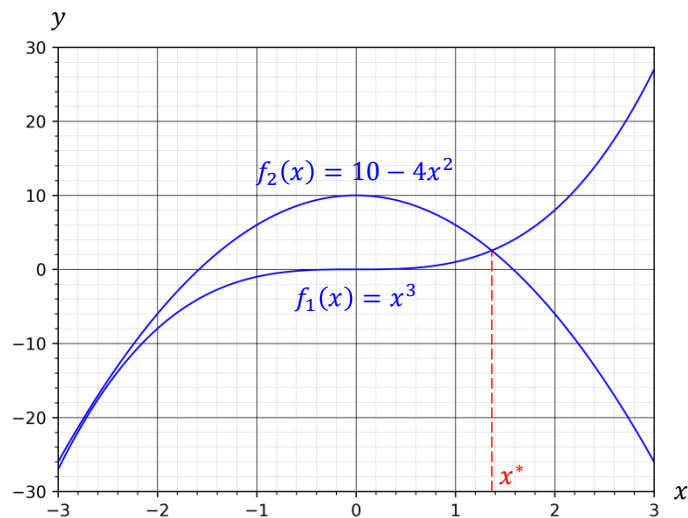
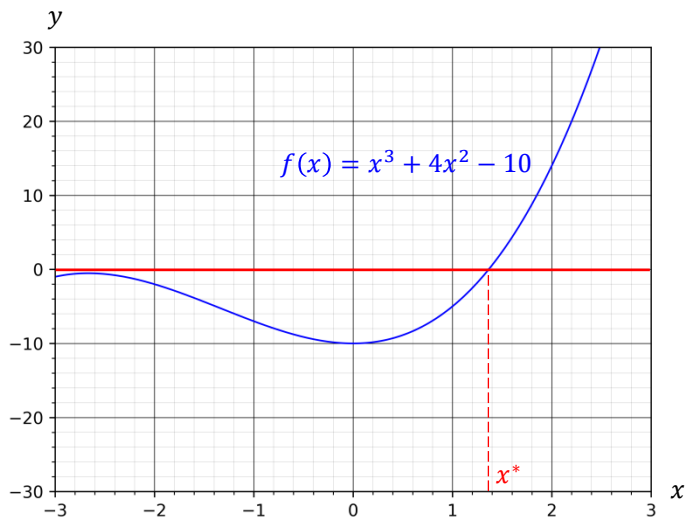
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Given: $x^3 + 4x^2 = 10 \rightarrow x^3 + 4x^2 - 10 = 0$

Bracketing Method

$$1 < x^* < 2$$

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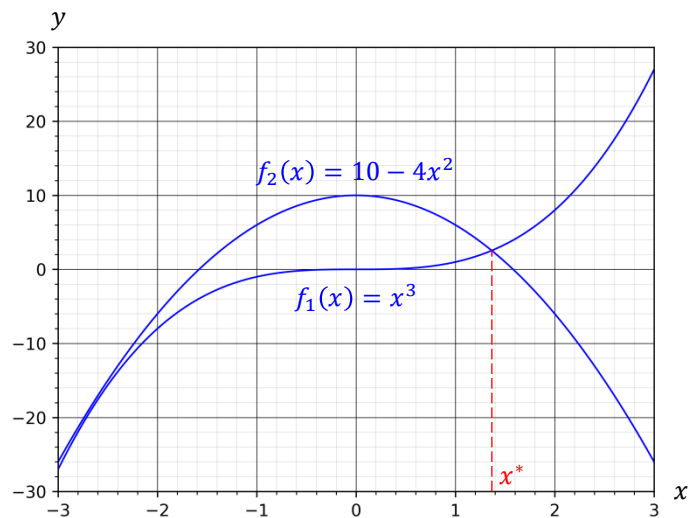
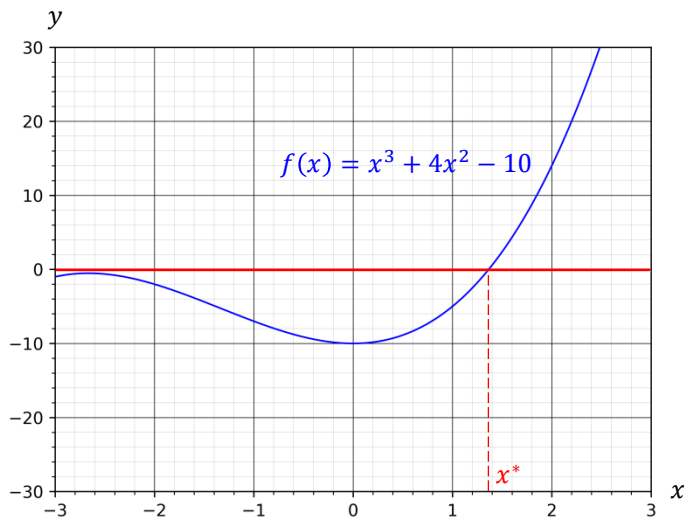
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Bracketing Method

$$\begin{matrix} 1 < x^* < 2 \\ \{a, b\} = \{0, 2\} \end{matrix}$$



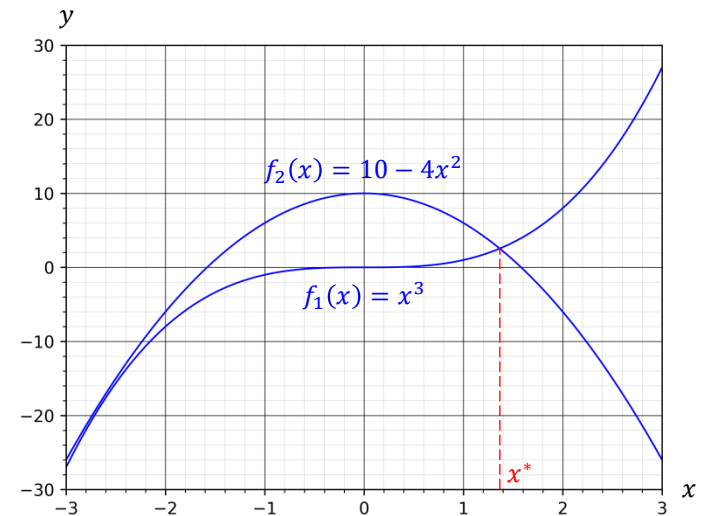
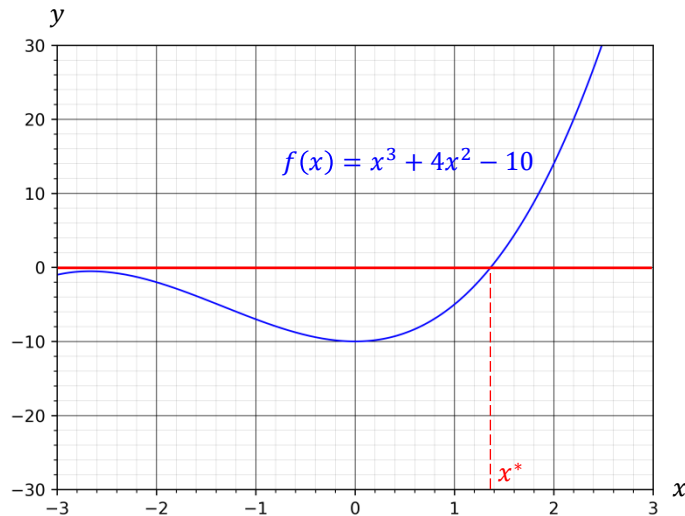
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Bracketing Method

$$\begin{aligned} 1 &< x^* < 2 \\ \{a, b\} &= \{0, 2\} \\ \{a, b\} &= \{1, 2\} \\ &\vdots \end{aligned}$$



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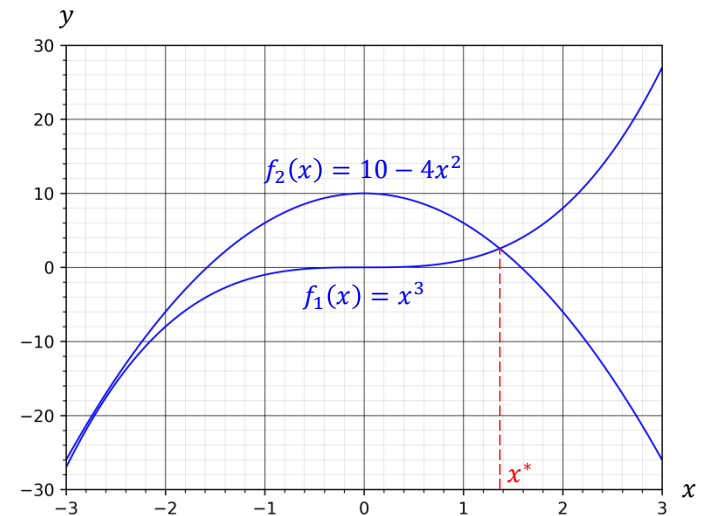
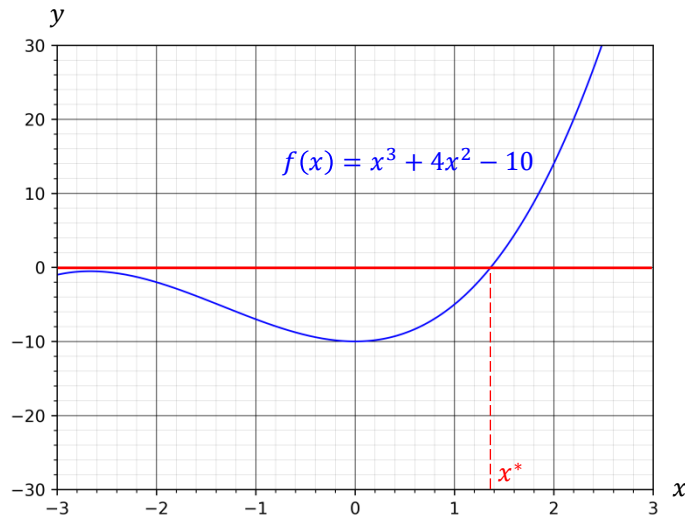
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Open Method

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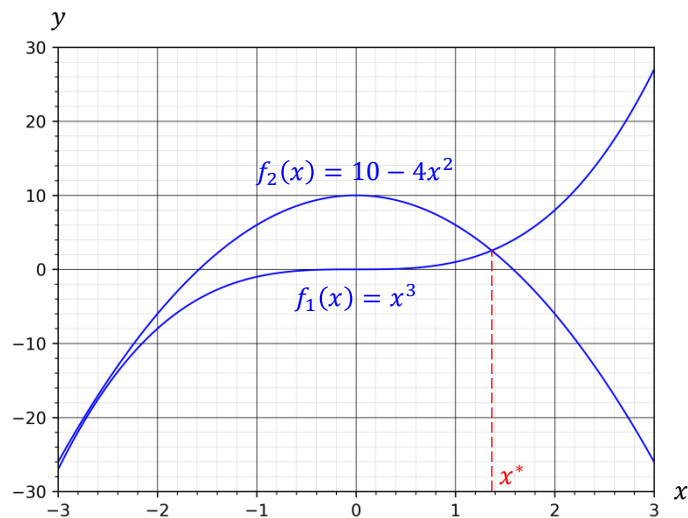
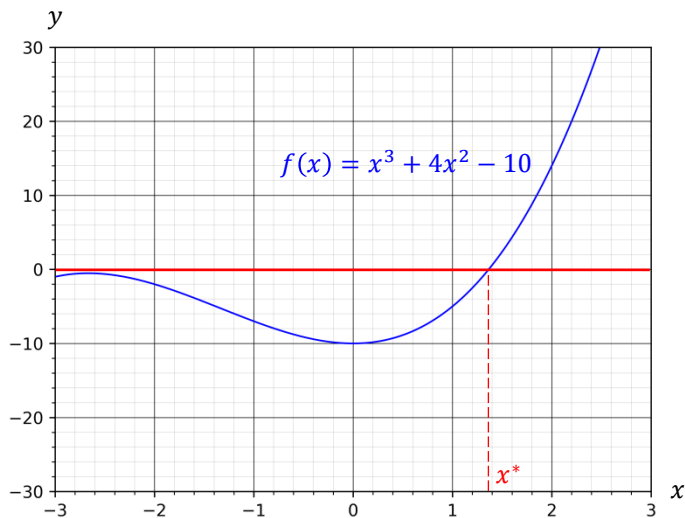
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Open Method

$$\begin{aligned} 1 &< x^* < 2 \\ x^{(0)} &= 1, 1.5, 2, \dots \end{aligned}$$



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Bracketing Method

$$\begin{aligned} 1 &< x^* < 2 \\ \{a, b\} &= \{0, 2\} \\ \{a, b\} &= \{1, 2\} \\ &\vdots \end{aligned}$$

Open Method

$$\begin{aligned} 1 &< x^* < 2 \\ x^{(0)} &= 1, 1.5, 2, \dots \\ \{x^{(-1)}, x^{(0)}\} &= \{0, 1\}, \{1, 2\}, \dots \\ &\vdots \end{aligned}$$

